SILENT ALIASING AND CERTIFIED SAMPLING DESIGN FOR FIXED FOURIER-FEATURE MODELS

Weiping Yan

University of Minnesota, Twin Cities, Minneapolis, MN, USA

ABSTRACT

Fixed Fourier-feature coordinate models—the linear core of Fourier-feature implicit neural representations—fit a signal by least squares on an exponential dictionary Λ. We analyze, and then *design against*, the aliasing of out-of-band content. (i) **Structured indistinguishability.** A visibility/aliasability calculus shows that *arbitrary nonuniform subsets of a sampling grid inherit the grid's aliasing equivalence classes*: an integer tone congruent mod Q to a model frequency is *exactly* indistinguishable from it—zero training residual, full-energy failure, and no estimator can avoid it. Irregular-on-a-grid sampling does *not* break exact aliasing. (ii) **Randomization breaks it, at a rate.** Under off-grid jitter the fold coherence has mean equal to the jitter characteristic function $\chi_\eta(2\pi rQ)$, so the squared visibility is $\approx 1 - |\chi_\eta(2\pi rQ)|^2$; i.i.d. sampling gives a finite-N, dictionary-referenced concentration bound on worst-case aliasability. (iii) **Certified design.** The anti-aliasing objective is *related to* a D_s-optimal (nuisance-parameter) / null-steering criterion on sample times (classical; credited); we give a fast surrogate that improves the aliasability–conditioning trade-off, and—our main added tool—a computable *continuum certificate*, a deterministic post-hoc guarantee that bounds worst-case aliasability over an entire band from the samples alone (for *any* realized design), which the finite-candidate concentration bound does not by itself provide. The certification machinery (gridding plus a Bernstein/Lipschitz slack) is standard; the contribution is its application to sample-only aliasability. On held-out frequencies near the concern set the design cuts function-space worst-case aliasability from 0.58 (random) to 0.15 ± 0.01 (vs. the exact D_s design 0.24 at matched optimization budget), degrading toward random outside the targeted band; the 1-D continuum certificate bounds it band-wide at 0.52 vs. 0.72 (random). Closed-form implementations are checked against independent simulations (proofs are separate); trained SIREN/MLP networks are an explicitly labeled empirical extension with failures reported.

Index Terms— Sampling theory, aliasing, identifiability, optimal experimental design, Fourier features.

1. INTRODUCTION

Coordinate networks with Fourier features are standard continuous signal models [1, 2, 3]. Their analytically exact core—the object of every theorem here—is the *fixed Fourier-feature coordinate model*: a finite frequency set $\Lambda = \{\omega_1, \dots, \omega_m\} \subset \mathbb{R}$ with linear coefficients, fit by least squares. Trained nonlinear networks enter only as an *empirical extension* (Sec. 6); adaptive/learned frequencies are out of scope—we make no claim for "all INRs".

A practitioner choosing *when* and *where* to sample faces three questions when the signal carries energy outside Λ. **Q1**: when is that energy exactly invisible in the samples, fitting perfectly yet corrupting the reconstruction? **Q2**: how does the sampling pattern change this? **Q3**: can one *design* the samples to prevent it, with a guarantee? This paper answers all three and closes the loop with a constructive, certified design.

Contributions. (1) A visibility/aliasability calculus and its central corrective (Thm. 1): arbitrary nonuniform *grid subsets* inherit the grid's aliasing equivalence classes, so irregular sampling on a grid does not break exact aliasing—with an estimator-independent two-point converse. (2) A finite-N, dictionary-referenced quantification of how off-grid randomization breaks the fold: a jitter characteristic-function law and an i.i.d. concentration bound (Thm. 2). (3) A *certified constructive design* (Sec. 5): we recognize the anti-aliasing objective as a D_s-optimal / null-steering criterion on sample times, give a fast joint surrogate that an ablation shows improves the tested aliasability–conditioning trade-off over each single-objective variant (at matched budget), and prove a computable *continuum aliasability certificate* that guarantees the worst case over an entire out-of-band interval from the samples only. (4) An honest empirical program: real-data both sides, a trained-network extension with its SIREN failure reported, and 2-D, all with confidence intervals and released failing runs.

Relation to prior work. Classical modulo-N aliasing [4] is the full-grid special case of Thm. 1, which covers arbitrary grid subsets; the generalized-sampling recast [5, 6] underlies our least-squares view. That randomized timing suppresses aliasing is classical—Balakrishnan's characteristic-function attenuation and alias-free random sampling [7, 8, 9]; those are asymptotic spectral-replica statements, whereas Thm. 2 gives finite-N, dictionary-referenced control (a column-norm of the generalized aliasing operator [10]), uniform over candidates. **Our design principle is not new and we do not claim it**: protecting a model block while nulling its coupling to a specified set is exactly D_s-optimal (nuisance-parameter) experimental design [11, 12] and LCMV/MVDR null-steering [13, 14]. Our contribution is the identifiability analysis that says *which* cross-terms to null and *why grid designs cannot*, together with the *continuum certificate* (Sec. 5)—a sample-only guarantee over a whole band, which the finite-candidate bound and random designs lack. Basis mismatch [15] is the complementary near-atom regime; the Lomb–Scargle window [16] predicts periodogram leakage. The structured-dictionary view of INRs [17], INR achievability theorems [18, 19, 20], an NTK missing-frequency floor [21], anti-aliasing INR architectures [22, 23, 24], and empirical SIREN aliasing [25] are adjacent; none gives the grid-inheritance corrective, the finite-N bound, or a certified design.

Code, data provenance, full proofs, and a claim ledger: `github.com/appleweiping/inr-aliasing-limits`. Portions of the code and text were produced with AI assistance under the author's direction (see the Compliance statement).

2. SETUP

Signals live on $[0,1)$; $e_\omega(x) = e^{i2\pi\omega x}$; $\mathcal{M}_\Lambda = \{\sum_k c_k e_{\omega_k}\}$ with Λ fixed. Samples $T = \{t_j\}_{j=1}^N$ give $S_T f = (f(t_j))_j$, the synthesis matrix $\boldsymbol{\Phi} \in \mathbb{C}^{N\times m}$ ($\boldsymbol{\Phi}_{jk} = e_{\omega_k}(t_j)$), and $\boldsymbol{y} = S_T f + \boldsymbol{\varepsilon}$. Write $f = f_{\text{in}} + f_{\text{out}}$, $f_{\text{out}} = \sum_l a_l e_{\nu_l}$, $\nu_l \notin \Lambda$. Ordinary least squares $\hat{\boldsymbol{c}} = \boldsymbol{\Phi}^\dagger \boldsymbol{y}$ is used throughout. For $\nu \in \mathbb{R}$, the *sample atom* $\boldsymbol{\phi}_\nu = (e^{i2\pi\nu t_j})_j$ defines the *visibility* and *aliasability*

$$v_T(\nu) = \tfrac{1}{\sqrt{N}}\|(\boldsymbol{I} - \boldsymbol{P}_\Lambda)\boldsymbol{\phi}_\nu\| \in [0,1], \qquad a_T(\nu) = \|\boldsymbol{\Phi}^\dagger \boldsymbol{\phi}_\nu\|, \quad (1)$$

$\boldsymbol{P}_\Lambda$ the projector onto $\operatorname{range}\boldsymbol{\Phi}$: v_T is the sample energy the model cannot absorb, and a_T is the *coefficient* bias a unit tone induces. Coefficient norm is signal energy only for orthonormal (integer) Λ; in general we use the *function-space* aliasability $a_{L^2,T}(\nu) = (\Delta c^* G_{L^2} \Delta c)^{1/2}$, $\Delta c = \boldsymbol{\Phi}^\dagger \boldsymbol{\phi}_\nu$, with the continuous Gram $G_{L^2}[k,\ell] = \prod_d e^{i\pi\delta_d} \operatorname{sinc}\delta_d$ ($\delta = \omega_\ell - \omega_k$; $= \boldsymbol{I}$ for integer frequencies) — this is the $L^2([0,1)^d)$ energy the aliased tone injects into the fit, and is the metric used throughout Sec. 5. Under white noise, $\|\hat{\boldsymbol{c}} - \boldsymbol{c}^\star\| \le \|\boldsymbol{\varepsilon}\|/\sigma_{\min}(\boldsymbol{\Phi})$ (standard; Lemma S1), and $\|\hat{f} - f\|_{L^2}^2$ splits exactly into a modeled-component (aliasing-bias+noise) term, a cross term, and a truncation term $\boldsymbol{a}^* G_{\text{out}} \boldsymbol{a}$ that *no* sampling removes (Prop. S1; Fig. 1d and Fig. 2d).

3. STRUCTURED INDISTINGUISHABILITY

Theorem 1 (Exact equivalence on grid subsets)**.** *Let $\Lambda \subset \mathbb{Z}$, T any subset of the rate-Q grid $\{0,\dots,Q-1\}/Q$ with $\sigma_{\min}(\boldsymbol{\Phi}) > 0$ ($N \ge m$; Λ distinct mod Q), $\nu \in \mathbb{Z}$. (i) $v_T(\nu) = 0$ iff $\boldsymbol{\phi}_\nu \in \operatorname{range}\boldsymbol{\Phi}$; if $\nu \equiv \omega_k \pmod Q$ then $\boldsymbol{\phi}_\nu = \boldsymbol{\phi}_{\omega_k}$* on every such subset*, so $v_T(\nu) = 0$, the fit is the unit coefficient on atom k, and the training residual is exactly zero. (ii) With $f_1 = f_{\text{in}} + a e_\nu$, $f_2 = f_{\text{in}} + a e_{\omega_k}$ ($\nu \equiv \omega_k$, $\nu \ne \omega_k$): $S_T f_1 = S_T f_2$, so the observation laws coincide under any noise and* any *estimator has $\max_i \|\hat{f} - f_i\|_{L^2} \ge |a|/\sqrt{2}$. (iii) If $\nu \not\equiv \omega_k$ for all k and $[\boldsymbol{\Phi}\ \boldsymbol{\phi}_\nu]$ has full column rank then $v_T(\nu) > 0$.*

Sketch. (i) $e^{i2\pi\nu g/Q}$ depends only on $\nu \bmod Q$; (ii) two-point (Le Cam) with $\|e_\nu - e_{\omega_k}\|_{L^2} = \sqrt{2}$; (iii) the rank condition (supplement). *Silent aliasing* is (i)–(ii): the samples are fit to the noise floor while the *least-squares* reconstruction error equals the full out-of-band energy ($\sqrt{2}|a|$) and *no* estimator does better than $|a|/\sqrt{2}$ (ii)—for *any* admissible grid-subset (parts (i)–(ii) need no rank condition; the full-rank claims of (iii) require $N \ge m$), with ν arbitrarily far from Λ. The corrective is (i)'s *on every such subset*: a common intuition holds that irregular sampling avoids aliasing; on a grid it does not, no matter how irregular the subset. Classical aliasing is $T =$ full grid [4].

4. RANDOMIZATION BREAKS IT, AT A RATE

Theorem 2 (Off-grid randomization)**.** *Let $\Lambda \subset \mathbb{Z}$, $|\Lambda| = m$. (a)* (jitter) *$t_j = g_j/Q + \eta_j$, η_j i.i.d. mean-zero with characteristic function χ_η, $\nu = \omega_k + rQ$ ($r \ne 0$). For arbitrary g_j the fold coherence $\frac{1}{N}\sum_j e^{i2\pi(\nu-\omega_k)t_j}$ has mean $\chi_\eta(2\pi rQ)$ and concentrates at rate $2\sqrt{\log(4/\delta)/N}$; if the g_j are drawn i.i.d. uniformly and Λ is distinct mod Q then $v_T(\nu)^2 = 1 - |\chi_\eta(2\pi rQ)|^2 + O_P(m\sqrt{\log(m/\delta)/N})$, giving (Gaussian σ_t) $v_T(\nu) \approx 2\pi|r|Q\sigma_t$. (b)* (i.i.d.) *If the limiting normalized Gram is $\succeq \lambda_{\min}\boldsymbol{I}$ and limiting model–candidate coherences vanish on a finite Ω, $|\Omega| = K$ ($\lambda_{\min} = 1$ for the uniform density and integer-disjoint candidates), then with $\varepsilon = 2\sqrt{\log(4(mK+m^2)/\delta)/N}$, $m\varepsilon < \lambda_{\min}$, w.p. $\ge 1-\delta$, $\max_{\nu\in\Omega} a_T(\nu) \le \sqrt{m}\,\varepsilon/(\lambda_{\min} - m\varepsilon)$.*

Sketch. (a) the on-grid fold phase is 1; jitter adds i.i.d. unit-modulus factors (empirical characteristic function; Hoeffding). (b) Hoeffding+union over $mK+m^2$ coherences, then a Neumann/perturbation bound on $a_T(\nu) = \|(\boldsymbol{\Phi}^*\boldsymbol{\Phi}/N)^{-1}(\boldsymbol{\Phi}^*\boldsymbol{\phi}_\nu/N)\|$. A variance-aware matrix-Bernstein/Chernoff refinement (supplement) tightens the sufficient budget $\sim 30\times$ and, in its data-dependent form, is finite at every full-rank realized design—including the acquisition budgets where the design is used—while an in-expectation *matching* lower bound shows the $N^{-1/2}$ rate is order-optimal (Fig. 1a,b). Two honest boundaries: on-grid designs never escape Thm. 1; and (b) controls only a *finite* Ω—the certificate of Sec. 5 lifts that to a whole band for a given design.

5. CERTIFIED ANTI-ALIASING SAMPLING DESIGN

Given Λ, a candidate out-of-band set Ω the practitioner wants visible, and a budget N, we choose sample times to make $\boldsymbol{\Phi}$ well conditioned *and* $\boldsymbol{\Phi}^*\boldsymbol{\Psi}_\Omega$ small ($\boldsymbol{\Psi}_\Omega$ the Ω atoms): jointly, high $\min_\nu v_T(\nu)$ and low function-space aliasability $\max_\nu a_{L^2,T}(\nu)$ at small $\kappa(\boldsymbol{\Phi})$. This asymmetric block target—protect $\boldsymbol{\Phi}^*\boldsymbol{\Phi} \to \boldsymbol{I}$, null $\boldsymbol{\Phi}^*\boldsymbol{\Psi}_\Omega \to 0$, ignore the Ω self-block—is *related to* D_s-optimal (nuisance-parameter) design [11, 12], whose exact objective is the Schur-complement log-determinant $\log\det(\boldsymbol{\Phi}^*\boldsymbol{\Phi} - \boldsymbol{\Phi}^*\boldsymbol{\Psi}_\Omega(\boldsymbol{\Psi}_\Omega^*\boldsymbol{\Psi}_\Omega)^{-1}\boldsymbol{\Psi}_\Omega^*\boldsymbol{\Phi})$ (supplement), and structurally to LCMV/MVDR null-steering [13, 14]. We credit those, do *not* claim the principle, and compare against the exact D_s design; our additions are the identifiability reading (which cross-terms, and why grid designs fail: Thm. 1) and the certificate.

Surrogate and trade-off. The exact objectives are non-smooth and (via $\boldsymbol{\Phi}^\dagger$) non-submodular, so the position-space greedy carries no approximation guarantee; but lifting the surrogate to the design *measure* convexifies it, and Frank–Wolfe then yields a *computable* ε-optimality certificate (the duality gap; supplement)—a guarantee, not merely the post-hoc band certificate. We minimize the smooth surrogate $J(T) = \frac{1}{N^2}\sum_{k,\nu}|(\boldsymbol{\Phi}^*\boldsymbol{\Psi}_\Omega)_{k\nu}|^2 + \beta\frac{1}{N^2}\sum_{k\ne l}|(\boldsymbol{\Phi}^*\boldsymbol{\Phi})_{kl}|^2$ (difference-frequency sums) by coordinate descent over continuous times (a grid-restricted greedy variant inherits Thm. 1 and cannot break exact folds). Sweeping β traces an aliasability–conditioning frontier (Fig. 2a) that improves on both the coherence-only and condition-only ablations and matches or beats the exact D_s and E-optimal designs at the same budget.

Continuum certificate. For a design T, $a_{L^2,T}(\nu)^2 = \boldsymbol{\phi}_\nu^*\boldsymbol{M}\boldsymbol{\phi}_\nu$ with $\boldsymbol{M} = \boldsymbol{\Phi}(\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1}G_{L^2}(\boldsymbol{\Phi}^*\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^*$ (or $G_{L^2} = \boldsymbol{I}$ for the coefficient norm) is a real *finite exponential polynomial* in ν (frequencies $t_l - t_j$, generally non-integer); by an explicit Lipschitz grid bound (a Bernstein-type inequality [26], $|g'| \le 2\pi\sum_{j,l}|\boldsymbol{M}_{jl}||t_l - t_j|$) a fine grid plus a slack term *certifies* $\max_{\nu\in[a,b]} a_{L^2,T}(\nu)$ over an entire band from the samples alone (Prop. 1; sound and tight in the tests; computed via an eigendecomposition, vacuous if $\boldsymbol{\Phi}$ is rank deficient). It is a deterministic *post-hoc* guarantee for *any* realized design—random, E-optimal, or optimized alike; AliasGuard's role is to make the certified value small. The finite-candidate bound Thm. 2(b) does not by itself yield such a continuum guarantee.

Proposition 1 (Certificate)**.** *For any T with $\sigma_{\min}(\boldsymbol{\Phi}) > 0$, $\max_{\nu\in[a,b]} a_{L^2,T}(\nu) \le \left(\max_{\text{grid}} \boldsymbol{\phi}^*\boldsymbol{M}\boldsymbol{\phi} + L\,h/2\right)^{1/2}$ with grid spacing h and $L = 2\pi\sum_{j,l}|\boldsymbol{M}_{jl}||t_l - t_j|$; an analogous bound lower-bounds $\min_{\nu\in[a,b]} v_T(\nu)$.*

Result. Over $\geq$ 20 paired outer scenarios (each its own dictionary—orthonormal and non-orthonormal—concern set, noise, and init), the design, evaluated on *held-out* frequencies it never saw, cuts function-space worst-case aliasability to $0.15{\pm}0.01$ near the concern set vs. 0.58 (random) and 0.24 (exact D_s); the advantage *degrades honestly* as the test band moves off target (shifted, scaled, or broadband: Fig. 2b). It persists across the budget sweep (Fig. 2c). A separated decomposition (Fig. 2d) shows the design lowers the out-of-band aliasing-bias term while the truncation floor and the estimation noise are unaffected; the designs' noise gains ($\sigma_{\min}^{-2}$, $\mathrm{tr}((\mathbf{\Phi}^*\mathbf{\Phi})^{-1})$) and condition numbers—distinct quantities—are reported alongside. Over a continuous 1-D band the design *certifies* $\max a_{L^2} \leq 0.52$ vs. 0.72 (random) and 0.62 (exact D_s), a non-vacuous (< 1) guarantee in 20 of 24 scenarios (the rest are reported vacuous, not hidden). A preliminary 2-D held-out-vector comparison gives nan vs. nan (random)—a small margin that is separated from random but not every baseline and uses only an i.i.d.-uniform baseline, so we do not present it as a headline. Honest scope: the tool applies where sample *times are chosen*—active sampling, acquisition design, or scheduling a coordinate network's training queries—not already-acquired data; it needs a focused concern set ($K = O(N)$), and for a broad band densely covered ($K \gg N$) or a badly misspecified set it loses its edge.

6. EXPERIMENTS

Closed-form implementations are checked against independent simulations (a CPU test suite run by the GitHub Actions continuous-integration workflow on every push); the proofs are given separately in the supplement. A separate integration job checks that every number in this paper, its claim ledger, and the README equals the committed result JSON. Those JSONs carry seeds, configs, full-SHA/dirty-flag provenance, and host/CPU/GPU stamps; every figure is regenerated from them. ("CI" here means that integration workflow; the statistical confidence intervals below are a distinct quantity.)

Theory (Fig. 1). (a) the jitter law matches $\sqrt{1-|\chi_\eta|^2}$ across two decades (the agreement is empirical and holds well beyond the regime where the finite-N union-bound linear law is provably tight), exact fold ($v \approx 10^{-14}$) at $\sigma_t{=}0$; (b) i.i.d. worst-case aliasability decays $N^{-1/2}$ beneath the bound; (c) the grid-subset coherent tone stays invisible at every N while i.i.d. sampling makes even the per-N adversarial candidate visible; (d) jitter removes the aliasing-bias term, never truncation.

Statistical protocol. The design study uses $\geq$ 20 *paired outer scenarios*; each fixes its own dictionary (half non-orthonormal), concern set, held-out set, noise, and optimizer initialization, and *every* method sees the identical scenarios. We report paired-bootstrap CIs over scenarios (the replication unit), match compute budget across methods (equal objective-evaluation/restart budgets; wall-clock recorded), and evaluate generalization on five held-out band types (near / omitted / shifted / scaled / broadband), never a single fixed offset.

Design (Fig. 2). The four panels above; the joint objective, held-out generalization, budget sweep, and signal-domain payoff.

Real signals, sample-only. On CO_2, sunspots, and nine speech segments (anti-aliased decimation; full records), sample-only Lomb–Scargle bandwidth selection attains RMSE 0.511 vs. an evaluation-only oracle 0.441 on CO_2; on broadband speech *no* selector beats RMSE$\approx$1.01—the truncation term dominates at that budget (the real-data face of the converse). Details, the detection dichotomy (a coherent fold is indistinguishable from its in-band twin by any sample-only test; for visible tones *only* the residual detector has a closed-form operating characteristic—the exact noncentral-χ^2 power curve—while the other detectors are empirical baselines shown for comparison), and a trained-network extension (FF-MLP fold match 75%, correlation 0.96 over $\geq$ 20 sampling scenarios with independently drawn feature/weight seeds and a separate weight-seed-stability control; the trained *SIREN* responses are *inconsistent with the initialization-NTK prediction*, correlation 0.47. We measured the init-vs-trained NTK drift to test whether this is a lazy-training artifact: it is *not*—the SIREN kernel drifts *less* than the FF-MLP kernel (0.64 vs. 6.92 median relative), yet the FF-MLP prediction still matches—so the fixed-feature/init-NTK description simply does not extrapolate to trained SIRENs, and we make no claim there) with 2-D results are in the supplement, with all failing runs released.

7. LIMITATIONS AND CONCLUSION

The theory covers fixed-feature linear-coefficient models; nonlinear networks are an empirical extension whose init-NTK fold predictor *fails for SIRENs*, and adaptive frequencies are open. A variance-aware matrix-Bernstein bound with a matching in-expectation lower bound now replaces the loose union bound; the design needs a focused concern set. Within that scope: irregular-on-a-grid sampling does not break exact aliasing (Thm. 1); off-grid randomization does, at an explicit finite-N rate (Thm. 2); and a D_s-optimal / null-steering design with a continuum certificate turns the analysis into a deployable, *certified* anti-aliasing sample pattern (the certificate is a genuine deterministic upper bound, not an optimality claim). Practical guidance: never sample purely on a grid when out-of-band content is possible; target a focused concern set; and certify the band, not a finite list.

Compliance with Ethical Standards. This is a theoretical and computational study; it used only publicly available data (Mauna Loa CO_2, NOAA; sunspot numbers, SILSO/Royal Observatory of Belgium; Open Speech Repository Harvard sentences), under those sources' terms, and involved no human or animal subjects. The authors declare no conflicts of interest and no external funding. Code and text were produced with AI assistance (Anthropic Claude) under the author's direction; all results were generated by the released, reproducible code.

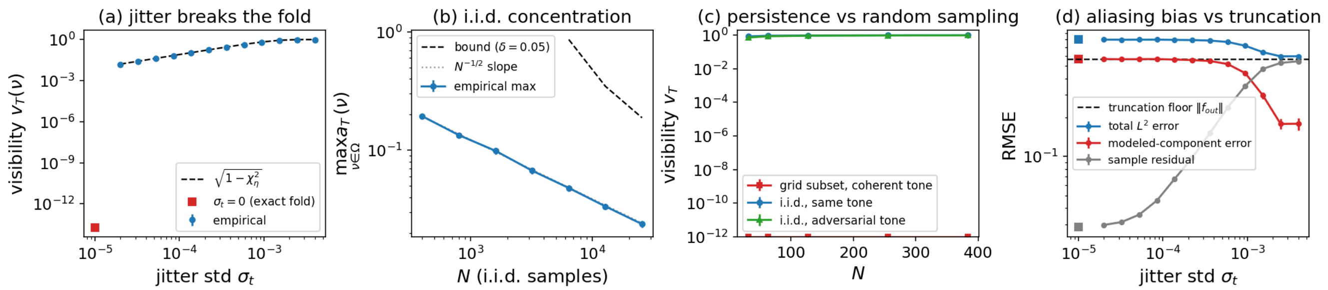

Fig. 1. Theory validation (mean $\pm$ 95% CI, 20 draws). (a) jitter law $\sqrt{1-|\chi_\eta|^2}$; (b) i.i.d. worst-case aliasability vs. the bound; (c) grid persistence vs. i.i.d. (fixed *and* adversarial tones); (d) jitter removes aliasing bias, not truncation.

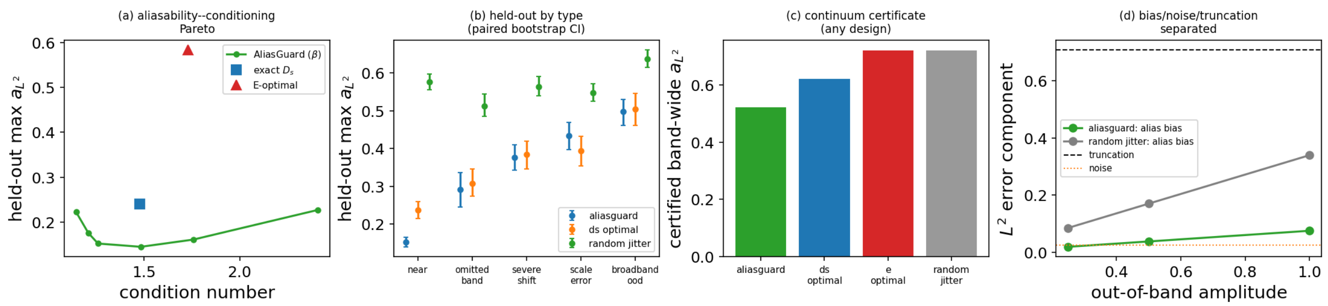

Fig. 2. Certified anti-aliasing design ($\geq$ 20 paired outer scenarios; function-space aliasability a_{L^2}). (a) aliasability–conditioning frontier: sweeping β traces the AliasGuard frontier, at/below the exact D_s and E-optimal points; (b) held-out aliasability by test type (paired-bootstrap CI)—strong near the concern set, degrading honestly off-target; (c) advantage across the sample budget; (d) the L^2 error separated into out-of-band aliasing bias, estimation noise, and truncation (only the bias term responds to the design).

8. REFERENCES

[1] V. Sitzmann, J. Martel, A. Bergman, D. Lindell, and G. Wetzstein, "Implicit neural representations with periodic activation functions," in *NeurIPS*, 2020.

[2] M. Tancik, P. Srinivasan, B. Mildenhall, et al., "Fourier features let networks learn high frequency functions in low dimensional domains," in *NeurIPS*, 2020.

[3] B. Mildenhall, P. Srinivasan, M. Tancik, et al., "NeRF: Representing scenes as neural radiance fields for view synthesis," in *ECCV*, 2020.

[4] C. E. Shannon, "Communication in the presence of noise," *Proc. IRE*, vol. 37, no. 1, pp. 10–21, 1949.

[5] B. Adcock and A. C. Hansen, "Generalized sampling and infinite-dimensional compressed sensing," *Found. Comput. Math.*, vol. 16, pp. 1263–1323, 2016.

[6] M. Unser, "Sampling—50 years after Shannon," *Proc. IEEE*, vol. 88, no. 4, pp. 569–587, 2000.

[7] A. V. Balakrishnan, "On the problem of time jitter in sampling," *IRE Trans. Inform. Theory*, vol. 8, no. 3, pp. 226–236, 1962.

[8] H. S. Shapiro and R. A. Silverman, "Alias-free sampling of random noise," *J. Soc. Indust. Appl. Math.*, vol. 8, no. 2, pp. 225–248, 1960.

[9] F. J. Beutler, "Alias-free randomly timed sampling of stochastic processes," *IEEE Trans. Inform. Theory*, vol. 16, no. 2, pp. 147–152, 1970.

[10] M. K. Transtrum, G. L. W. Hart, T. J. Jarvis, and J. P. Whitehead, "Generalized aliasing explains double descent and informs model design," *Phys. Rev. Research*, vol. 7, no. 4, pp. 043268, 2025, arXiv:2408.08294.

[11] S. D. Silvey, *Optimal Design*, Chapman & Hall, 1980.

[12] A. C. Atkinson, A. N. Donev, and R. D. Tobias, *Optimum Experimental Designs, with SAS*, Oxford Univ. Press, 2007.

[13] O. L. Frost, "An algorithm for linearly constrained adaptive array processing," *Proc. IEEE*, vol. 60, no. 8, pp. 926–935, 1972.

[14] H. L. Van Trees, *Optimum Array Processing*, Wiley, 2002.

[15] Y. Chi, L. L. Scharf, A. Pezeshki, and A. R. Calderbank, "Sensitivity to basis mismatch in compressed sensing," *IEEE Trans. Signal Process.*, vol. 59, no. 5, pp. 2182–2195, 2011.

[16] J. T. VanderPlas, "Understanding the Lomb–Scargle periodogram," *Astrophys. J. Suppl. Ser.*, vol. 236, no. 1, pp. 16, 2018.

[17] G. Yüce, G. Ortiz-Jiménez, B. Besbinar, and P. Frossard, "A structured dictionary perspective on implicit neural representations," in *CVPR*, 2022.

[18] M. Najaf and G. Ongie, "Towards a sampling theory for implicit neural representations," in *Asilomar Conf. Signals, Syst., Comput.*, 2024, arXiv:2405.18410.

[19] M. Najaf and G. Ongie, "Sampling theory for super-resolution with implicit neural representations," *SIAM J. Imaging Sci.*, 2025, arXiv:2506.09949.

[20] H. Saratchandran, S. Ramasinghe, V. Shevchenko, A. Long, and S. Lucey, "A sampling theory perspective on activations for implicit neural representations," in *ICML*, 2024, arXiv:2402.05427.

[21] M. Ma, Q. Zhu, Y. Zhan, Z. Yin, H. Wang, and Y. Zheng, "Robustifying Fourier features embeddings for implicit neural representations," *arXiv:2502.05482*, 2025.

[22] D. B. Lindell, D. Van Veen, J. J. Park, and G. Wetzstein, "BACON: Band-limited coordinate networks for multiscale scene representation," in *CVPR*, 2022.

[23] J. Barron, B. Mildenhall, M. Tancik, et al., "Mip-NeRF: A multiscale representation for anti-aliasing neural radiance fields," in *ICCV*, 2021.

[24] A. Shabanov, S. Govindarajan, C. Reading, et al., "BANF: Band-limited neural fields for levels of detail reconstruction," in *CVPR*, 2024.

[25] H. Chandravamsi, D. V. Shenoy, I. Zinn, Z. Chen, S. Pisnoy, and S. H. Frankel, "Spectral bottleneck in sinusoidal representation networks: Noise is all you need," *arXiv:2509.09719*, 2025.

[26] R. A. DeVore and G. G. Lorentz, *Constructive Approximation*, Springer, 1993, Bernstein inequality for trigonometric polynomials.